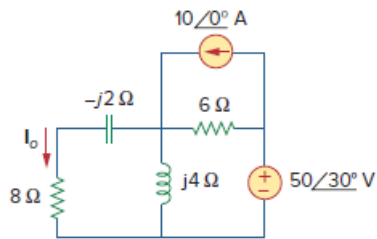


Problem

Find current I_o in the circuit of Fig. 10.8 using the superposition theorem.

Figure 10.8:



Step-by-step solution

Step 1 of 9

Consider the following circuit diagram:

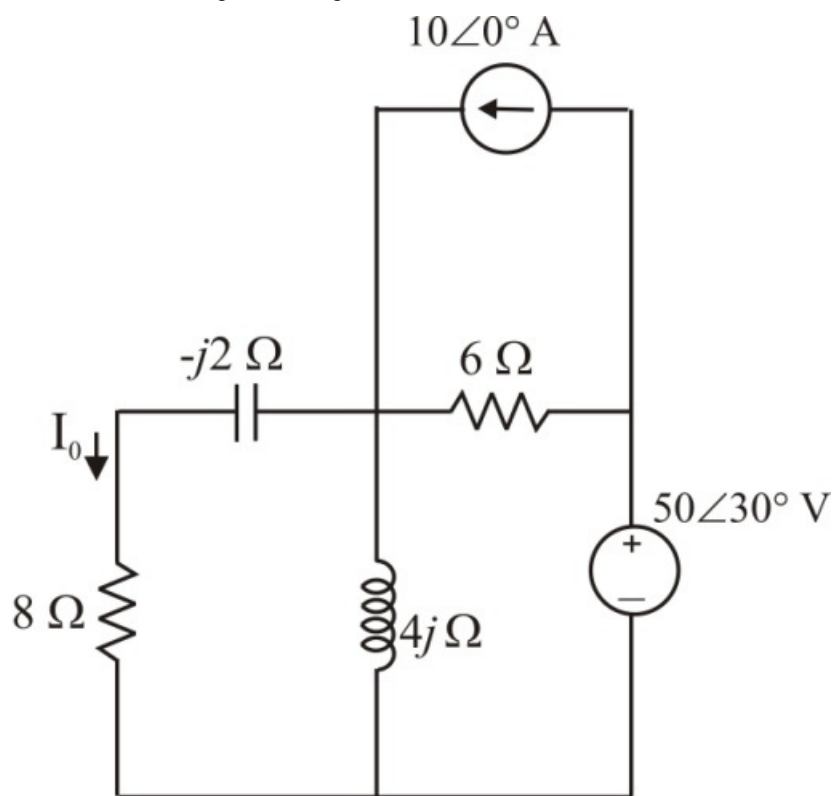


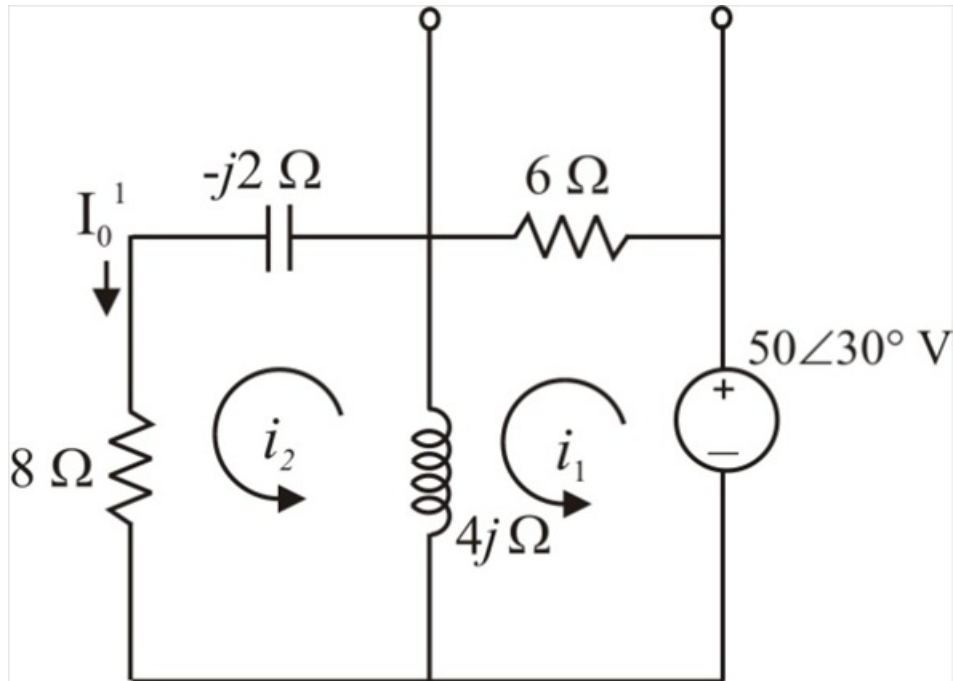
Figure 1

Step 2 of 9

Let us assume that $\mathbf{I}_o = \mathbf{I}'_o + \mathbf{I}''_o$.

Whereas, according to superposition theorem $\mathbf{I}'_o$ and $\mathbf{I}''_o$ are the currents due to voltage and current sources respectively.

Consider the following circuit in which current source is kept open circuit to find $\mathbf{I}'_o$.



Step 3 of 9

Figure 2

Step 4 of 9

Assume that the currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are flowing in the circuit.

Apply Kirchhoff's voltage law to mesh 1.

$$j4(\mathbf{I}_1 - \mathbf{I}_2) + 6(\mathbf{I}_1) = 50\angle 30^\circ$$

$$(6 + j4)\mathbf{I}_1 - j4\mathbf{I}_2 = 43.30 + j25 \dots\dots (1)$$

Apply Kirchhoff's voltage law to mesh 2.

$$8\mathbf{I}_2 - j2\mathbf{I}_2 + j4(\mathbf{I}_2 - \mathbf{I}_1) = 0$$

$$-j2\mathbf{I}_1 + (4 + j)\mathbf{I}_2 = 0 \dots\dots (2)$$

Step 5 of 9

Write the equations (1) and (2) in the matrix form.

$$\begin{bmatrix} 6+j4 & -j4 \\ -j2 & 4+j \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 43.30+j25 \\ 0 \end{bmatrix}$$

Calculate the determinant values of the matrices to find $\mathbf{I}_1$ and $\mathbf{I}_2$.

$$\begin{aligned} \Delta &= \begin{vmatrix} 6+j4 & -j4 \\ -j2 & 4+j \end{vmatrix} \\ &= (6+j4)(4+j) - (-j4)(-j2) \\ &= 28+j22 \\ \Delta_2 &= \begin{vmatrix} 6+j4 & 43.30+j25 \\ -j2 & 0 \end{vmatrix} \\ &= (0)(6+j4) - (j2)(43.30+j25) \\ &= -50+j86.60 \end{aligned}$$

Step 6 of 9

Calculate the current $\mathbf{I}'_o$ in mesh 2:

$$\begin{aligned} \mathbf{I}'_o &= \mathbf{I}_2 \\ &= \frac{\Delta_2}{\Delta} \\ &= \frac{-50+j86.60}{28+j22} \\ &= 0.398+j2.779 \end{aligned}$$

Therefore the value of current $\mathbf{I}'_o$ is $2.808\angle 81.84^\circ \text{ A}$.

Consider the following circuit in which voltage source is kept as short circuit to find $\mathbf{I}''_o$:

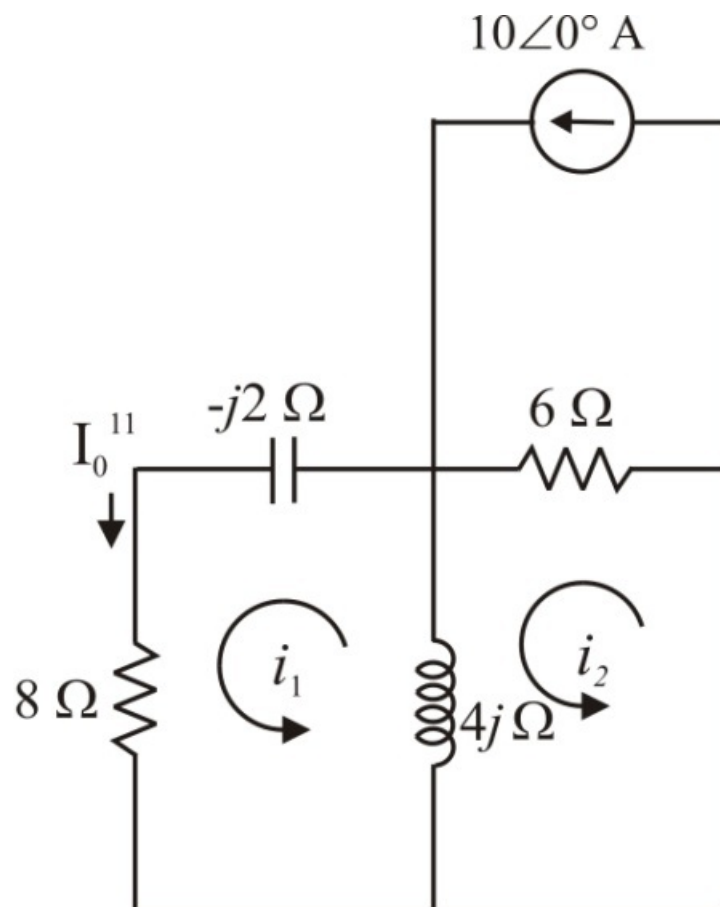


Figure 3

Step 7 of 9

Assume that the currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are flowing in the circuit.

Apply Kirchhoff's voltage law to mesh 1.

$$8\mathbf{I}_1 - j2\mathbf{I}_1 + j4(\mathbf{I}_1 - \mathbf{I}_2) = 0$$

$$\mathbf{I}_1(4 + j) - j2\mathbf{I}_2 = 0 \dots\dots (3)$$

Apply Kirchhoff's voltage law to mesh 2.

$$j4(\mathbf{I}_2 - \mathbf{I}_1) + 6(\mathbf{I}_2 - 10) = 0$$

$$-j4\mathbf{I}_1 + (6 + j4)\mathbf{I}_2 - 60 = 0$$

$$-j4\mathbf{I}_1 + (6 + j4)\mathbf{I}_2 = 60 \dots\dots (4)$$

Step 8 of 9

Write the equations (3) and (4) in the matrix form.

$$\begin{bmatrix} 4+j & -j2 \\ -j4 & 6+j4 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 60 \end{bmatrix}$$

Calculate the determinant values of the matrices to find $\mathbf{I}_1$ and $\mathbf{I}_2$.

$$\begin{aligned} \Delta &= \begin{vmatrix} 4+j & -j2 \\ -j4 & 6+j4 \end{vmatrix} \\ &= (4+j)(6+j4) - (-j2)(-j4) \\ &= (24 + j16 + j6 - 4) + 8 \\ &= 28 + j22 \end{aligned}$$

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 0 & -j2 \\ 60 & 6+j4 \end{vmatrix} \\ &= 0(6+j4) - (-j2)(60) \\ &= j120 \end{aligned}$$

Calculate the current $\mathbf{I}_o''$ in mesh 1:

$$\begin{aligned} \mathbf{I}_o'' &= \mathbf{I}_1 \\ &= \frac{\Delta_1}{\Delta} \\ &= \frac{j120}{28 + j22} \\ &= 2.08 + j2.649 \end{aligned}$$

Therefore the value of current $\mathbf{I}_o''$ is $3.369 \angle 51.84^\circ \text{ A}$.

Step 9 of 9

Therefore calculate the value of required current $\mathbf{I}_o$:

$$\begin{aligned} \mathbf{I}_o &= \mathbf{I}_o' + \mathbf{I}_o'' \\ &= 2.808 \angle 81.84^\circ + 3.369 \angle 51.84^\circ \\ &= 5.969 \angle 65.44^\circ \text{ A} \end{aligned}$$

Hence, the current value is calculated by using the superposition theorem and the value is

$$\boxed{5.969 \angle 65.44^\circ \text{ A}}.$$